

Derbyshire Shared Care Pathology Guidelines

Coeliac Disease

Purpose of Guideline

- When and how to investigate patients for Coeliac Disease
- What the results mean
- When and how to refer patients
- Monitoring of patients with Coeliac Disease

Definition

Coeliac disease is an autoimmune disease triggered by gluten in genetically susceptible individuals, leading to chronic inflammation of the small intestine with villous atrophy and flattening of the mucosa.

Who should be investigated for Coeliac Disease?

The prevalence of Coeliac Disease in Europe is thought to be about 1% with the majority of diagnoses made in adult life. In patients with gastrointestinal symptoms, the prevalence is approximately 5%. Prevalence is also increased in patients with type 1 diabetes (up to 7.7%), some autoimmune conditions (liver disease, thyroid disorders, pulmonary disease, ulcerative colitis) and IgA deficiency. Patients with confirmed iron deficiency anaemia and with suspected irritable bowel syndrome should have coeliac serology checked in line with national guidance (see references 1,2).

Classical presentation:

- Weight loss
- Chronic diarrhoea
- Steatorrhoea

Consider Coeliac Disease in patients with:

- Iron Deficiency Anaemia
- Folate Deficiency
- Hypocalcaemia/ Vitamin D deficiency
- Reduced bone density
- Lethargy
- Abdominal distension/bloating and other features of irritable bowel
- Arthralgia
- Infertility
- Dermatitis Herpetiformis
- Constipation

What tests should I request to investigate for Coeliac Disease?

First line

Anti-tissue transglutaminase antibody (tTG). Patients must have been on a diet containing adequate dietary gluten (equivalent to 4 slices of bread daily) for 6 weeks prior to the test. Negative results do not exclude Coeliac Disease if the patient has significantly reduced their wheat intake.

(Note: if total IgA has not previously been measured in a patient, the lab will automatically add this to a request for tTG in order to detect falsely low tTG results due to IgA deficiency. If a raised IgA is found, then depending on the clinical details the lab will add IgG, IgM and serum electrophoresis to significantly raised IgA (≥ 8 g/L).

Second line

Measurement of IgA-Endomysial Antibody (EMA) is requested by the laboratory as a second line test as outlined below. EMA analysis is more time-consuming and expensive than tTG and is therefore not carried out as a first-line test.

What do the results mean?

The flow chart on page 3 summarises the interpretation of tTG and EMA results. More detailed information is given in the subsequent pages.

What should I do if the result suggests Coeliac Disease?

South Derbyshire

- All adult patients with tTG >45 U/mL should be referred directly to the dietetic service and diagnosis confirmed by satisfactory response to gluten-free diet (see ref 3 and 5). Referral for consultant opinion is available if required by GP.
- All other adult patients should be referred to a consultant gastroenterologist for assessment. In most cases the consultant will refer the patient directly for endoscopy for confirmation of diagnosis by duodenal biopsy. Patients with positive biopsy results are then referred to the dietetic service and are not seen in the outpatient clinic.
- Paediatric patients should be referred to Paediatric gastroenterology for full assessment prior to considering changes to the diet.

North Derbyshire

- All adult patients with tTG >45 U/mL should be referred to Gastroenterology and diagnosis confirmed by satisfactory response to gluten-free diet (see ref 3 and 5). Referral for consultant opinion is available if required by GP.
- All other adult patients should be kept on a gluten containing diet and be referred to a consultant gastroenterologist for assessment. In most cases the consultant will refer the patient directly for endoscopy for confirmation of diagnosis by duodenal biopsy prior to review in clinic..
- Paediatric patients Should be referred according to tTG result before considering changes to diet.
- For tTG results < 100 U/mL refer to Gastroenterology at Sheffield Children's Hospital
For tGG results ≥ 100 U/mL refer to Paediatrics at Chesterfield Royal Hospital

Adult patients will usually be seen by the dietitian at diagnosis, They will then be discharged back to their GP for annual reviews unless the patient requires a short period of ongoing dietetic management, following which they will be discharged back to the for annual

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reviews. The following blood tests would be taken at each clinic appointment and then annually:

Full blood count

Serum Ferritin

Serum Folate

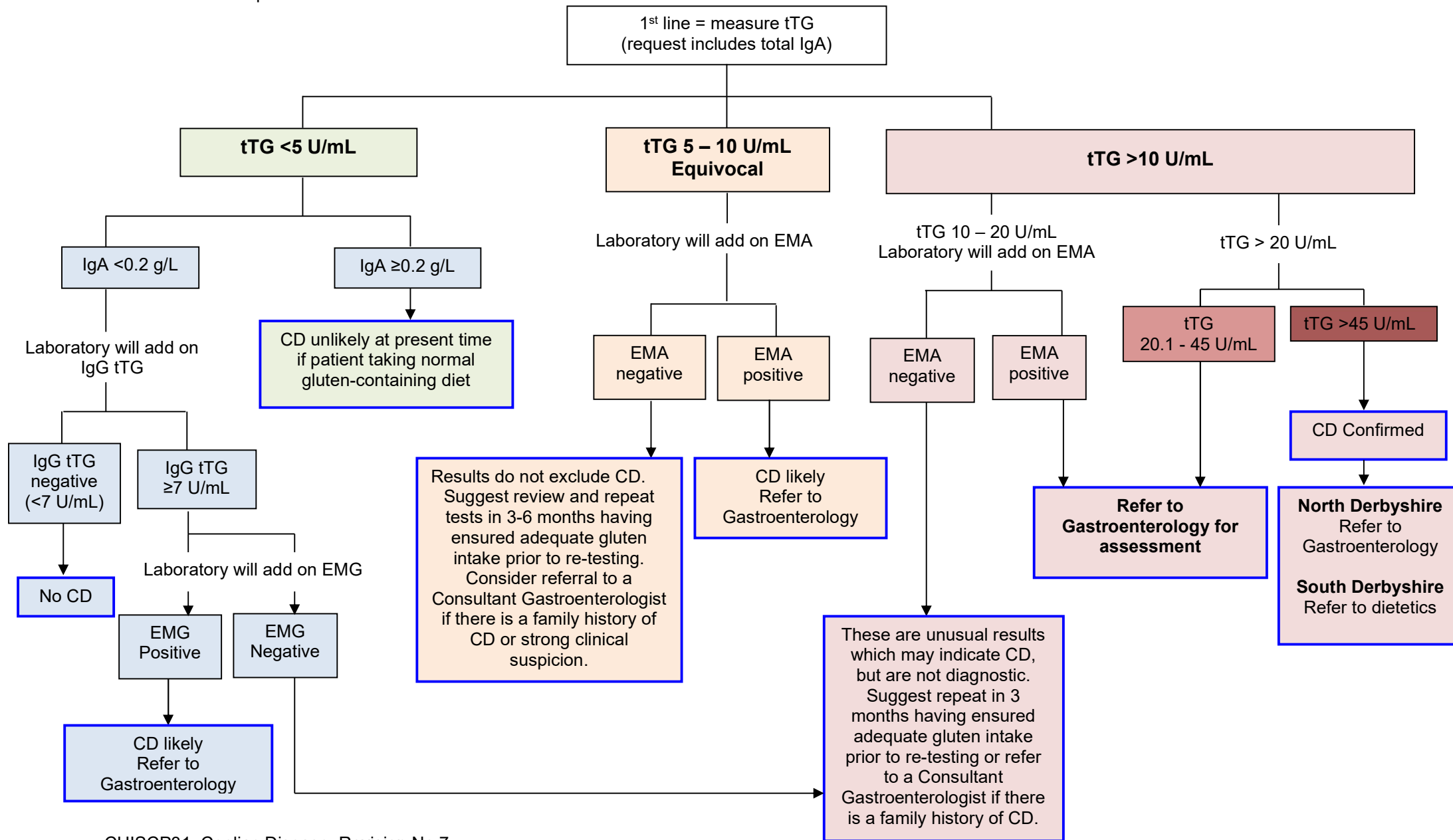
Vitamin B12

Liver function tests and serum calcium

Vitamin D

Coeliac Serology (tTG)

Adherence to a gluten-free diet can be monitored by measuring tTG. It takes approximately 4.5 weeks for the tTG to fall by 50% once a gluten free diet has been started.



Low tTG (≤ 0.3 U/mL)

- If patient has low total IgA levels (<0.2 g/L), the laboratory will test for IgG Anti tissue transglutaminase antibodies (IgG tTG) to exclude Coeliac Disease
- Positive IgG tTG results (≥ 7 U/mL) will be confirmed by IgG Endomysial antibodies (EMA) as a second line test
- Positive IgG tTG/EMA results in the presence of selective IgA deficiency are strongly suggestive of Coeliac Disease. Referral to a Consultant Gastroenterologist for further assessment is suggested.
- Selective IgA Deficiency is a relatively common (~ 1 in 700), and usually asymptomatic disorder

Normal tTG (0.4 - 4.9 U/mL)

- Coeliac disease is unlikely at the time of testing (sensitivity 96%), provided the patient is taking a normal gluten containing diet.

Equivocal tTG (5 - 10 U/mL)

The laboratory will request IgA-Endomysial Antibody (EMA) when tTG is 5 - 10 U/mL.

- **Equivocal tTG (5 - 10 U/mL), negative EMA, normal or raised total IgA**
These results do not exclude coeliac disease.
Suggest review and repeat tests in 3-6 months having ensured adequate gluten intake prior to re-testing. Consider referral to a Consultant Gastroenterologist if there is a family history of Coeliac Disease or strong clinical suspicion.
- **Equivocal tTG (5 - 10 U/mL), positive EMA**
These results are suggestive of Coeliac Disease and referral to a Consultant Gastroenterologist is suggested.

Raised tTG (>10 U/mL)

The laboratory will request IgA-Endomysial Antibody (EMA) when tTG is 10 - 20 U/mL.
This marker has a specificity of 99% for Coeliac Disease.

- **Raised tTG (10 – 20 U/mL), positive EMA**
These results are strongly suggestive of Coeliac Disease and referral to a Consultant Gastroenterologist is suggested.
- **Raised tTG (10 – 20 U/mL), negative EMA**
These are unusual results which may indicate Coeliac Disease, but are not diagnostic. Suggest repeat in 3 months having ensured adequate gluten intake prior to re-testing or refer to a Consultant Gastroenterologist if there is a family history of Coeliac Disease.
- **Raised tTG (20.1 - 45 U/mL)**
EMA is not necessary. Patients should be referred to a consultant gastroenterologist.
- **Raised tTG (>45 U/mL)**
EMA is not necessary. Patients should be referred to the dietetic clinic (South Derbyshire) or Nurse Practitioner (North Derbyshire).

References

1. Goddard AF, James MW, McIntyre AS and Scott BB. British Society of Gastroenterology Guidelines for the management of iron deficiency anaemia. Gut 2011;**60**:1309-1316
2. NICE guideline NG20: Coeliac Disease: Recognition, assessment and management (Sept 2015)
3. Hill PG and Holmes GKT. Coeliac Disease: a biopsy is not always necessary for diagnosis. Alimentary Pharmacology & Therapeutics 2008 ; 27(7): 572-7
4. Hill PG and McMillan SA. Anti-tissue transglutaminase antibodies and their role in the investigation of coeliac disease. Ann Clin Biochem 2006; 43: 105-117
5. Holmes GKT, Forsyth JM, Knowles S, Seddon H, Hill PG, Austin AS. Coeliac disease: further evidence that biopsy is not always necessary for diagnosis Eur J Gastroenterol Hepatol. 2017 Jun;29(6):640-64

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